

Chapter 9 Test model

- **70 Multiple Choice Questions (MCQs)**
 - **30 True/False Questions**
 - **Difficulty:** 70% Hard, 30% Very Hard
 - **Format:** Clear numbering, blank line (___) between each question
 - **Answer keys** are provided **after all questions**
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Multiple Choice Questions (1–70)

1. In ER-to-relational mapping, what is the primary key of a relation created from a regular entity type?
 - a. Any attribute of the entity
 - b. A surrogate key
 - c. One of the key attributes of the entity
 - d. The foreign key from a related entity
-
2. When mapping a weak entity type, the primary key of the resulting relation consists of:
 - a. Only the partial key of the weak entity
 - b. The primary key of the owner entity only
 - c. The combination of the owner's primary key and the weak entity's partial key
 - d. A newly generated surrogate key
-
3. Which of the following is NOT a valid approach for mapping a binary 1:1 relationship?
 - a. Foreign key in one of the two relations
 - b. Merging both entity types into a single relation
 - c. Creating a separate relation with both primary keys as foreign keys
 - d. Adding a multivalued attribute to one entity
-
4. In mapping a 1:N relationship, where is the foreign key placed?
 - a. In the relation corresponding to the '1' side
 - b. In the relation corresponding to the 'N' side
 - c. In a new relationship relation
 - d. In both relations

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5. For an M:N relationship, the standard mapping technique requires:
- Adding a foreign key to one entity relation
 - Merging both entities
 - Creating a new relation with foreign keys from both entities
 - Using a type attribute
-

6. The primary key of a relationship relation for an M:N binary relationship is:
- The primary key of the first entity
 - The primary key of the second entity
 - A surrogate key
 - The combination of both foreign keys
-

7. How is a multivalued attribute mapped in the relational model?
- As a simple attribute in the same relation
 - As a separate relation with the entity's primary key as a foreign key
 - By converting it into a derived attribute
 - By ignoring it during mapping
-

8. The primary key of the relation created for a multivalued attribute includes:
- Only the multivalued attribute
 - Only the foreign key from the owning entity
 - Both the multivalued attribute and the foreign key
 - A surrogate key
-

9. For an n-ary relationship ($n > 2$), the mapping produces:
- A single merged relation of all participating entities
 - A new relation with foreign keys from all participating entity relations
 - A foreign key added to the largest entity relation
 - No relation—n-ary relationships cannot be mapped
-

10. Which mapping option for specialization requires that every entity in the superclass belongs to at least one subclass?
- Option 8A

- b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

11. In Option 8A for mapping specialization, the subclass relations contain:
- a. Only subclass-specific attributes
 - b. All superclass attributes plus subclass attributes
 - c. Only the primary key of the superclass
 - d. The primary key and subclass-specific attributes
-

12. Option 8C uses what kind of attribute to indicate subclass membership?
- a. Surrogate key
 - b. Boolean flag
 - c. Type (discriminating) attribute
 - d. Partial key
-

13. Option 8D is most suitable for which type of specialization?
- a. Total and disjoint
 - b. Partial and disjoint
 - c. Overlapping (non-disjoint)
 - d. Attribute-defined only
-

14. In mapping a shared subclass (multiple inheritance), what must be true about the superclasses?
- a. They must have the same primary key
 - b. They must be weak entities
 - c. They must participate in a 1:1 relationship
 - d. They must have disjoint attributes
-

15. What is used to map a category (union type) when the superclasses have different keys?
- a. Composite key
 - b. Partial key
 - c. Surrogate key
 - d. Type attribute

16. In the COMPANY database example, the DEPENDENT relation's primary key is:

- a. SSN
- b. DEPENDENT_NAME
- c. {ESSN, DEPENDENT_NAME}
- d. {SSN, RELATIONSHIP}

17. In Step 4 of the ER-to-relational mapping, attributes of the 1:N relationship itself are placed in:

- a. The '1' side relation
- b. The 'N' side relation
- c. A new relationship relation
- d. Both relations

18. Which of the following relationships in the COMPANY schema is M:N?

- a. MANAGES
- b. WORKS_FOR
- c. SUPERVISION
- d. WORKS_ON

19. The relation DEPT_LOCATIONS is created due to which ER construct?

- a. Weak entity
- b. Multivalued attribute
- c. M:N relationship
- d. Composite attribute

20. In Option 8B, the superclass relation is:

- a. Created with all attributes
- b. Not created at all
- c. Created only with the key
- d. Merged into a view

21. What is a major limitation of the relational model regarding ER constraints?

- a. It cannot represent entities

- b. It cannot enforce maximum cardinality ratios like 1:10
 - c. It cannot store string data
 - d. It cannot handle foreign keys
-

22. Which mapping strategy minimizes null values in specialization hierarchies when subclasses are disjoint and total?
- a. Option 8A
 - b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

23. In Figure 9.7 (mapping categories), the OWNER relation uses:
- a. Composite key from BANK and PERSON
 - b. Surrogate key OwnerId
 - c. Boolean flags
 - d. Type attribute
-

24. The foreign key in the EMPLOYEE relation representing WORKS_FOR is named:
- a. DNUMBER
 - b. DEPT_NO
 - c. DNO
 - d. MANAGER_SSN
-

25. Which of the following is a goal of the ER-to-relational mapping algorithm?
- a. Maximize redundancy
 - b. Preserve all information and minimize nulls
 - c. Eliminate all foreign keys
 - d. Convert all attributes to multivalued
-

26. In Step 3 (1:1 mapping), why is it better to place the foreign key in the entity with total participation?
- a. To avoid nulls
 - b. To reduce relations
 - c. To simplify queries
 - d. To enforce disjointness

27. For a weak entity with no partial key, the primary key of its relation is:

- a. A surrogate key
- b. The owner's primary key alone
- c. Not defined
- d. A composite of all attributes

28. The relationship relation for SUPPLY (ternary) has a primary key consisting of:

- a. One foreign key
- b. Two foreign keys
- c. Three foreign keys
- d. A surrogate key

29. Which of the following is NOT preserved in relational mapping?

- a. Entity attributes
- b. Participation constraints (total/partial)
- c. Key constraints
- d. Exhaustive classification in specialization

30. In Option 8C, if a specialization is overlapping, what issue arises?

- a. The type attribute can only hold one value
- b. Nulls increase
- c. Foreign keys conflict
- d. Primary key becomes composite

31. What distinguishes a category from a shared subclass?

- a. Categories have same-key superclasses
- b. Shared subclasses have different-key superclasses
- c. Categories have different-key superclasses
- d. Categories cannot have attributes

32. In mapping EER models, which option allows an entity to belong to multiple subclasses simultaneously?

- a. Option 8A

- b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

33. The MANAGES relationship in COMPANY is mapped using which approach?
- a. Merged relation
 - b. Foreign key in DEPARTMENT
 - c. Foreign key in EMPLOYEE
 - d. Separate relationship relation
-

34. If a multivalued composite attribute exists, how is it mapped?
- a. Ignored
 - b. As a single attribute
 - c. As simple components in a new relation
 - d. As a derived attribute
-

35. Which step handles attributes like {Phone1, Phone2, Phone3} if modeled as multivalued?
- a. Step 1
 - b. Step 2
 - c. Step 6
 - d. Step 5
-

36. In Step 7, for an n-ary relationship, simple attributes of the relationship are:
- a. Dropped
 - b. Added to one entity relation
 - c. Included in the new relationship relation
 - d. Converted to foreign keys
-

37. What is the main drawback of Option 8C for overlapping specializations?
- a. Wasted storage
 - b. Inability to represent multiple subclass membership
 - c. Complex joins
 - d. Loss of key constraints

38. In the relational schema, how is total participation in a 1:N relationship enforced?

- a. By NOT NULL on foreign key
- b. By unique constraint
- c. It cannot be fully enforced
- d. By check constraint on type

39. Which of the following is mapped to a relation with a composite primary key by default?

- a. Regular entity
- b. Weak entity
- c. 1:1 relationship with total participation
- d. Multivalued attribute

40. The attribute HOURS in WORKS_ON comes from:

- a. EMPLOYEE
- b. PROJECT
- c. The WORKS_ON relationship type
- d. A derived calculation

41. In Option 8D, how many Boolean flags are needed for 4 subclasses?

- a. 1
- b. 2
- c. 4
- d. 0

42. A shared subclass like STUDENT_ASSISTANT inherits:

- a. Different keys from each superclass
- b. The same key from all superclasses
- c. A new surrogate key
- d. No key

43. Which mapping option is most storage-efficient for a total, disjoint specialization?

- a. Option 8A
- b. Option 8B

- c. Option 8C
 - d. Option 8D
-

44. In Figure 9.5(c), the type attribute is named:

- a. JobType
 - b. EmployeeType
 - c. SubclassID
 - d. Category
-

45. The foreign key in DEPENDENT referencing EMPLOYEE is named:

- a. SSN
 - b. ESSN
 - c. EMP_ID
 - d. DEP_SSN
-

46. What is the primary reason to avoid the "relationship relation" approach for 1:N relationships?

- a. It increases nulls
 - b. It's redundant and inefficient
 - c. It violates normalization
 - d. It cannot store attributes
-

47. In Step 1, composite keys are handled by:

- a. Creating a surrogate key
 - b. Using all simple components as part of the primary key
 - c. Dropping non-key components
 - d. Converting to multivalued
-

48. Which construct requires a new relation even if it has no attributes?

- a. Weak entity
 - b. M:N relationship
 - c. 1:1 relationship
 - d. Multivalued attribute
-

49. The DEPT_LOCATIONS relation has which primary key?

- a. DLOCATION
 - b. DNUMBER
 - c. {DNUMBER, DLOCATION}
 - d. LOC_ID
-

50. In EER mapping, if a superclass has 1000 entities but only 100 belong to any subclass, which option minimizes nulls?

- a. Option 8A
 - b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

51. Which of the following cannot be directly enforced in the relational model after ER mapping?

- a. Key constraints
 - b. Referential integrity
 - c. Disjointness constraint in specialization
 - d. Domain constraints
-

52. In Option 8A, how is referential integrity between superclass and subclass maintained?

- a. Foreign key from subclass to superclass
 - b. Same primary key value
 - c. Trigger
 - d. View
-

53. The main advantage of Option 8D over 8C is:

- a. Less storage
 - b. Supports overlapping subclasses
 - c. Simpler queries
 - d. No nulls
-

54. In the SUPPLY ternary relationship, if one of the participating entities has a composite key, how is it handled in the relationship relation?

- a. Only one part is used

- b. All simple components are included as foreign keys
 - c. Replaced by surrogate
 - d. Ignored
-

55. Which step maps a relationship like "Teaches" between Professor, Course, and Semester (3-way)?
- a. Step 4
 - b. Step 5
 - c. Step 7
 - d. Step 3
-

56. If an entity has a derived attribute, how is it treated in relational mapping?
- a. Stored as a regular attribute
 - b. Not mapped (computed at query time)
 - c. Mapped as multivalued
 - d. Requires a new relation
-

57. In mapping a 1:1 relationship with partial participation on both sides, which approach is safest?
- a. Merge into one relation
 - b. Foreign key in either relation
 - c. Separate relationship relation
 - d. Ignore the relationship
-

58. What is the biggest risk of using Option 8B for a partial specialization?
- a. Data duplication
 - b. Loss of superclass entities not in any subclass
 - c. Null values
 - d. Key conflicts
-

59. In Figure 9.7, the STUDENT_ASSISTANT relation includes keys from:
- a. STUDENT and EMPLOYEE
 - b. GRADUATE and EMPLOYEE
 - c. PERSON and DEPARTMENT
 - d. Only STUDENT

60. Categories are mapped similarly to:

- a. Weak entities
 - b. Strong entities with surrogate keys
 - c. M:N relationships
 - d. Multivalued attributes
-

61. Which of the following is true about the foreign key in a weak entity relation?

- a. It can be null
 - b. It must match the owner's primary key and cannot be null
 - c. It is optional
 - d. It is always composite
-

62. In Step 6, if a multivalued attribute is also composite, the new relation includes:

- a. Only the composite name
 - b. The simple components of the composite attribute
 - c. A surrogate key only
 - d. Nothing—it's not mappable
-

63. The mapping algorithm prioritizes:

- a. Minimizing relations
 - b. Preserving information and minimizing nulls
 - c. Eliminating foreign keys
 - d. Using surrogate keys everywhere
-

64. In Option 8C, what happens if an entity belongs to two subclasses in an overlapping hierarchy?

- a. Two tuples are created
 - b. Only one subclass is recorded
 - c. An error occurs
 - d. Boolean flags are used instead
-

65. Which ER construct is the only one that always produces a relation with a composite primary key made of foreign keys?

- a. Weak entity
 - b. 1:1 relationship
 - c. M:N relationship
 - d. Multivalued attribute
-

66. In the COMPANY schema, the SUPERVISION relationship is:

- a. M:N
 - b. 1:N
 - c. 1:1
 - d. Ternary
-

67. The attribute DNO in EMPLOYEE is a:

- a. Primary key
 - b. Partial key
 - c. Foreign key
 - d. Derived attribute
-

68. For a weak entity with a composite partial key, the primary key of its relation is:

- a. The owner's key only
 - b. The partial key only
 - c. Owner's key + all simple parts of partial key
 - d. Surrogate key
-

69. In Step 3, the "merged relation" option is best when:

- a. Participation is partial on both sides
 - b. Both participations are total
 - c. One side is weak
 - d. The relationship has attributes
-

70. Which of the following is a valid reason to use a surrogate key in category mapping?

- a. Superclasses have different primary keys
- b. To reduce storage
- c. To avoid composite keys
- d. All of the above

True/False Questions (71–100)

71. In ER-to-relational mapping, every relationship type must result in a new relation.

- A. True
 - B. False
-

72. A weak entity's relation always includes the owner's primary key as a foreign key.

- A. True
 - B. False
-

73. Option 8B can be used for partial specializations.

- A. True
 - B. False
-

74. Multivalued attributes are stored in the same relation as the entity in the relational model.

- A. True
 - B. False
-

75. The primary key of an M:N relationship relation is always a surrogate key.

- A. True
 - B. False
-

76. In Option 8C, the type attribute can represent multiple subclass memberships.

- A. True
 - B. False
-

77. Categories require a surrogate key only when superclasses have different keys.

- A. True
 - B. False
-

78. The relational model can fully enforce total participation constraints from ER diagrams.

- A. True
 - B. False
-

79. In Step 4, the foreign key is placed on the '1' side of a 1:N relationship.

- A. True
 - B. False
-

80. A shared subclass must have the same key attribute across all its superclasses.

- A. True
 - B. False
-

81. Option 8D uses a single type attribute to indicate subclass membership.

- A. True
 - B. False
-

82. The `DEPENDENT_NAME` in the `DEPENDENT` relation is a partial key.

- A. True
 - B. False
-

83. All attributes of a superclass are duplicated in every subclass relation in Option 8A.

- A. True
 - B. False
-

84. N-ary relationships are mapped by creating a new relation with foreign keys from all participating entities.

- A. True
 - B. False
-

85. The `MANAGES` relationship in `COMPANY` is 1:N.

- A. True

B. False

86. In Option 8B, no relation is created for the superclass.

A. True

B. False

87. A derived attribute is always stored in the relational schema.

A. True

B. False

88. The goal of minimizing null values is always achievable in ER-to-relational mapping.

A. True

B. False

89. Composite attributes are stored as a single field in the relational model.

A. True

B. False

90. In Step 6, the primary key of the new relation for a multivalued attribute includes the entity's primary key and the attribute value.

A. True

B. False

91. Overlapping specializations can be correctly represented using Option 8C.

A. True

B. False

92. The WORKS_ON relation has a primary key of {ESSN, PNO}.

A. True

B. False

93. In Step 3, the foreign key approach for 1:1 relationships can lead to nulls if participation is partial.

- A. True
 - B. False
-

94. A category is a type of specialization.

- A. True
 - B. False
-

95. The relational model can preserve exhaustive classification constraints from EER diagrams.

- A. True
 - B. False
-

96. In Option 8A, subclass relations have the same primary key as the superclass.

- A. True
 - B. False
-

97. The SUPPLY relationship relation's primary key includes SNAME, PARTNO, and PROJNAME.

- A. True
 - B. False
-

98. Weak entities can exist without an owner entity in the relational model.

- A. True
 - B. False
-

99. All relationship attributes in 1:N relationships are placed in the 'N' side relation.

- A. True
 - B. False
-

100. Step 7 applies only to ternary relationships, not higher n-ary ones.

- A. True
 - B. False
-

Answer Key

Multiple Choice (1–70)

- 1. c
- 2. c
- 3. d
- 4. b
- 5. c
- 6. d
- 7. b
- 8. c
- 9. b
- 10. b
- 11. d
- 12. c
- 13. c
- 14. a
- 15. c
- 16. c
- 17. b
- 18. d
- 19. b
- 20. b
- 21. b
- 22. b
- 23. b
- 24. c
- 25. b
- 26. a
- 27. b
- 28. c
- 29. d
- 30. a
- 31. c

32. d

33. b

34. c

35. c

36. c

37. b

38. c

39. b

40. c

41. c

42. b

43. b

44. a

45. b

46. b

47. b

48. b

49. c

50. a

51. c

52. b

53. b

54. b

55. c

56. b

57. c

58. b

59. a

60. b

61. b

62. b

63. b

64. b

65. c

66. b

67. c

68. c

69. b

70. a

True/False (71–100)

- 71. B (False) — only M:N and n-ary always create new relations
 - 72. A (True)
 - 73. B (False) — only for total
 - 74. B (False) — requires separate relation
 - 75. B (False) — it's the combo of foreign keys
 - 76. B (False) — only one value allowed
 - 77. A (True)
 - 78. B (False) — only partially via NOT NULL
 - 79. B (False) — on the 'N' side
 - 80. A (True)
 - 81. B (False) — that's Option 8C; 8D uses multiple Booleans
 - 82. A (True)
 - 83. B (False) — only the key is in subclass relations in 8A
 - 84. A (True)
 - 85. B (False) — it's 1:1
 - 86. A (True)
 - 87. B (False) — usually not stored
 - 88. B (False) — not always possible
 - 89. B (False) — broken into simple attributes
 - 90. A (True)
 - 91. B (False) — cannot represent multiple membership
 - 92. A (True)
 - 93. A (True)
 - 94. B (False) — category is union, not specialization
 - 95. B (False) — cannot enforce exhaustive classification
 - 96. A (True)
 - 97. A (True)
 - 98. B (False) — must reference owner
 - 99. A (True)
 - 100. B (False) — applies to any $n > 2$
-